List of things we need for the report

Introduction, data & methods (approaches, pre-processing, etc), results & analysis of those results, discussions & conclusion, appendix with full code

Introduction

Traumatic Brain Injury (TBI) is an acute event, and its effects can continue long after the initial injury. This project examines whether TBI is associated with an ongoing process of structural brain degeneration. We focus especially on the left hippocampus, a region closely tied to memory and cognitive functioning and one that is known to be vulnerable after brain injury. For this reason, Left Hippocampus Total Volume (HPC_L_TOT) is used as the main outcome variable in the analysis.

Our work was structured around three related core points. First, we ask whether individuals with TBI show a faster decline in hippocampal volume over time than healthy controls. Second, within the injured group, we explore whether exposure to cognitive therapy is associated with differences in the rate of structural decline. Third, we examine whether preserving brain structure necessarily corresponds to better clinical functioning by studying the relationship between hippocampal volume and cognitive quality-of-life scores. Along with the hippocampal volume, we also consider the ventricle-to-brain ratio (VBR) as a supporting structural marker, since ventricular expansion can reflect loss of surrounding brain tissue. Together, these analyses allow us to move from structural change to possible recovery-related factors to functional outcomes.

Data & Methods

We started our work with an Excel dataset that had repeated brain measurements for both injured patients and healthy controls. The data are first imported in wide form and then converted into longitudinal long format using the tolong() R function from spida2, so that each row represents one observation for one subject at one time point. Several key variables are then converted into numeric form, including HPC_L_TOT, VBR, tothrsCOG (total hours of cognitive therapy), and LAQ_cogtot (cognitive quality-of-life score).

Pre-processing and cohort construction

The dataset is then split into two groups: injured patients and controls. Injured patients are identified as subjects whose IDs do not begin with "c", while controls are identified by IDs that do begin with "c".

As a side tangent, in both groups, observations are restricted to time >= 2, and only subjects with at least two usable observations are kept so that longitudinal change can be meaningfully estimated. A single brain scan provides a snapshot of a patient's current volume, but it tells us nothing about the changes that we are trying to measure. To calculate a slope, we removed any subjects with less than two points. This allowed us to determine the direction and speed of the injury effects. After adjusting for patients with more than two scans, we actually found that all patients met this threshold prior to adjustment! In this case, it was interesting to highlight as many studies may have empty subjects or missing values.

For the injured group, time is defined relative to the date of injury (DOI). The code converts both the scan date and DOI into date format, then calculates:

$$time\ in\ years = \frac{(date - DOI)}{365.25}$$

This creates a continuous measure of years since injury. This is because different individuals were injured on different days, and so it keeps a standardized time.

For controls, who do not have an injury date, time is instead defined relative to each subject's earliest observed scan date, giving a measure of years since baseline observation. Since healthy controls lack a DOI, we identify each subject's earliest available scan as their personal baseline using the *aggregate* function, then merge this back into the main dataset. In simpler terms, the control patients "DOI" is their earliest scan.

$$time_years = \frac{date - date_base}{365.25}$$

After removing missing values and selecting the main variables of interest, the injured and control datasets are combined into one master dataset with group labels. The final analysis dataset includes subject ID, time in years, hippocampal volume, group, age, sex, VBR, therapy hours, and cognitive outcome score.

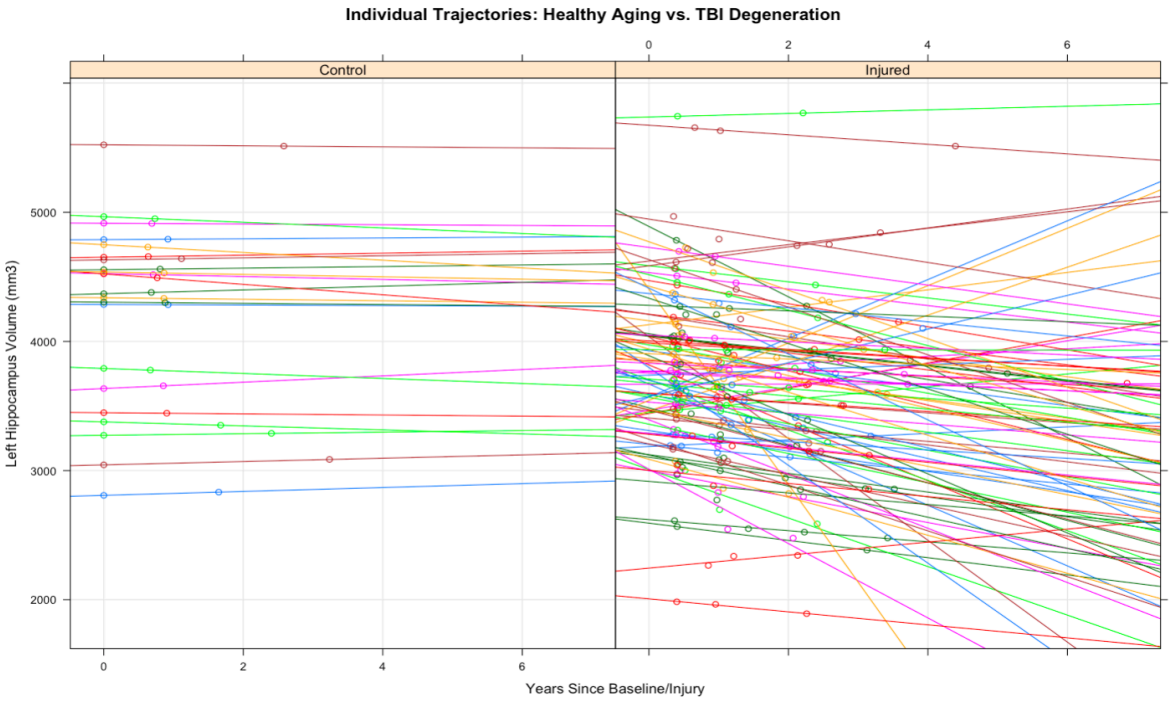
Analytical Approach - High-Level Summary

Note: A detailed description of each result is in the next segment of this paper.

The main statistical approach is linear mixed-effects modeling, which is appropriate because the data contain repeated measurements from the same individuals over time. Mixed models allow the analysis to estimate both the overall group-level trends and subject-specific differences in baseline volume and rate of change. In the code (which can be referred to in the appendix section), this is done by including a random intercept and random slope for time within each subject.

Structural degeneration analysis

To test whether TBI is associated with accelerated hippocampal decline, the code first generates individual trajectory plots of HPC_L_TOT over time for controls and injured patients. This contrasts the individual brain volume paths of healthy controls against TBI patients. As we expected, the healthy individuals HPC_L_TOT over time stayed relatively flat while the injured individuals HPC_L_TOT over time generally decreased. This hinted to us that the injury is also a catalyst for an ongoing process of structural loss.



It then fits a baseline mixed model with hippocampal volume as the response and time, group, and their interaction as predictors. The interaction term is the key quantity because it tests whether the slope of hippocampal change over time differs between the injured and control groups. An adjusted version of this model is then fit by adding age and sex as covariates, since both may influence brain volume independently of injury status.

Table: Adjusted LMM Results: Age & Sex Controlled

	Value	Std.Error	DF	t-value	p-value
(Intercept)	4196.2515	302.3607	171	13.8783	0.0000
time_years	1.6320	29.2959	171	0.0557	0.9556
GroupInjured	-377.0127	156.0253	114	-2.4164	0.0173
age	-3.0159	3.8268	171	-0.7881	0.4317
sex	56.0233	129.1580	114	0.4338	0.6653
time_years:GroupInjured	-64.8783	30.7896	171	-2.1072	0.0366

From the results, we could say that, on average, a healthy control starts with about 4196mm^3 of hippocampal tissue. Also, that injured patient's HPC_L_TOT volume differs from the controls by 377mm^3 (indicating the initial trauma results in a big volume drop). The smoking gun however is the coefficient time_years:GroupInjured. Here, we found that TBI patients lose an extra ~65mm^3 of HPC_L_TOT a year when compared to the control group (with a p-value of 0.03 - it is statistically significant).

The code also uses the fitted fixed effects from this adjusted model to create a simple prediction table showing projected hippocampal volume trajectories for controls and injured patients at 0, 2, 5, and 10 years. Here, we used the regression formula to calculate the projected volume at 4 different points.

Table: The Progressive Toll of TBI: Cumulative Volume Loss

Year	Healthy Vol	Injured Vol	Volume Gap	% Total Loss
0	4196.3	3819.2	377.0	9.0
2	4199.5	3692.7	506.8	12.1
5	4204.4	3503.0	701.4	16.7
10	4212.6	3186.8	1025.8	24.4

From the table we were able to see that initially, TBI patients on average lost about 9% of the HPC_L_TOT volume. From there, after 5 and 10 years respectively, there is a projected 17% and 24% volume loss over time compared to the healthy patients. Clearly this can potentially lead to significant long-term cognitive effects. It's important to note that this table is both a projection and using the average data points.

To strengthen the biological interpretation, the code also analyzes VBR using a similar mixed model. Subject-level slopes from the hippocampal model and the VBR model are then compared. This is meant to check whether subjects who lose hippocampal tissue more rapidly also tend to show greater ventricular expansion, which would be consistent with progressive structural loss. A small DAG is also included to justify the adjustment for age and sex as confounders in the structural analysis.

Recovery-factor analysis

The second part of the project focuses only on the injured group. After restricting to patients with available therapy information, the code fits another mixed-effects model with HPC_L_TOT as the outcome and time, total hours of cognitive therapy, and their interaction as predictors, while again controlling for age and sex. The interaction between time and therapy hours is used to explore whether greater therapy exposure is associated with a different rate of hippocampal change over time. This section is framed as an exploratory analysis of a possible recovery-related factor.

If cognitive therapy mitigates structural decline as a way of treating it, then we had to look at the interaction between time and total cognitive therapy hours with a linear mixed-effects model (see Table 4.). We found that the overall negative impact of time on hippocampal volume is still large($\beta=-64.38$, $p<0.0000$), and the interaction term between time and the therapy hours is not that significant ($\beta=0.1057$, $p=0.9553$). Although we saw a modest positive trend in the coefficient of therapy hours that indicates slightly larger time for therapy can be a recovery factor, we lacked the evidence that more therapy hours slow down brain volume loss for patients.

Table 4: Recovery LMM: Protective Trend of Cognitive Therapy

	Value	Std.Error	DF	t-value	p-value
(Intercept)	3762.7540	298.8714	135	12.5899	0.0000
time_years	-64.3844	15.2302	135	-4.2274	0.0000
tothrsCOG	0.1584	11.4384	87	0.0138	0.9890
age	-2.3562	4.3586	135	-0.5406	0.5897
sex	73.1697	156.3777	87	0.4679	0.6410
time_years:tothrsCOG	0.1057	1.8833	135	0.0561	0.9553

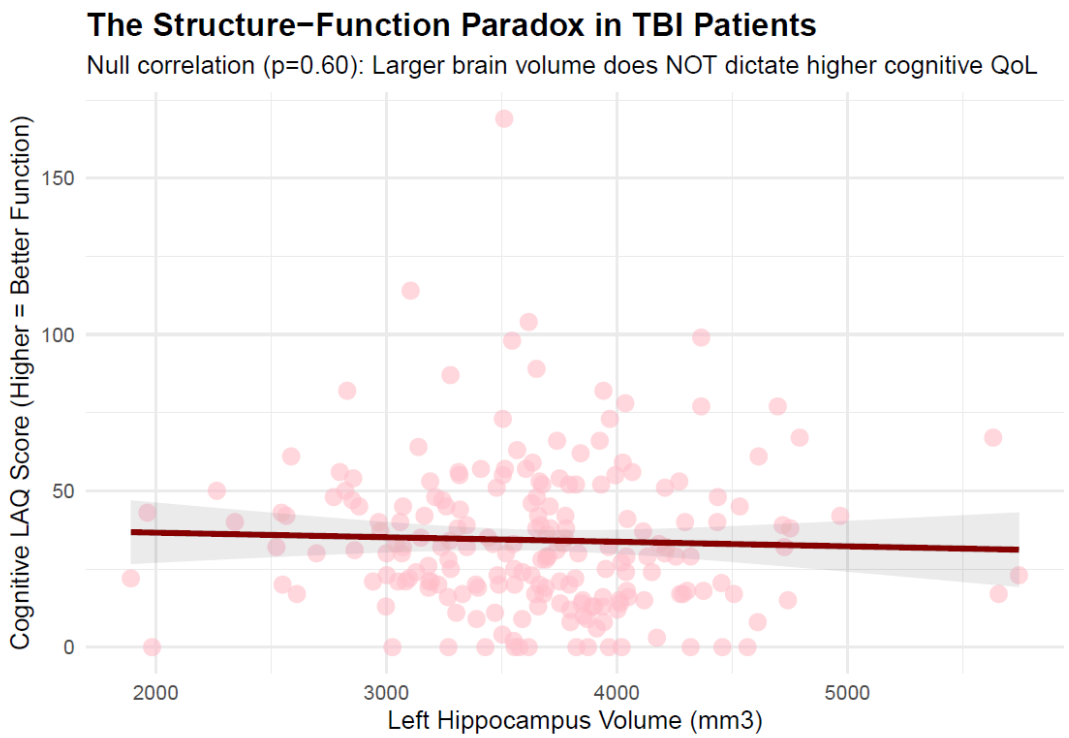
Clinical outcomes analysis

The final section asks whether structural preservation is directly tied to better functioning. For injured patients with non-missing LAQ_cogtot scores, the code creates a scatterplot of hippocampal volume versus cognitive quality-of-life score and overlays a simple linear regression line. A standard linear model is then fit with LAQ_cogtot as the response and HPC_L_TOT as the predictor. This part of the analysis is intended to test whether a larger preserved hippocampal volume is associated with better-reported cognitive functioning.

As seen in the scatter plot (See table and figure below.). and confirmed with the linear model, there is almost a flat trendline and zero correlation between brain volume and the LAQ score ($p=0.6003$). This means that brain volume alone will certainly not lead to better cognitive ability after injury. It is our best understanding that recovery from brain injury will likely involve other factors such as network adaptation rather than structure or amount of volume.

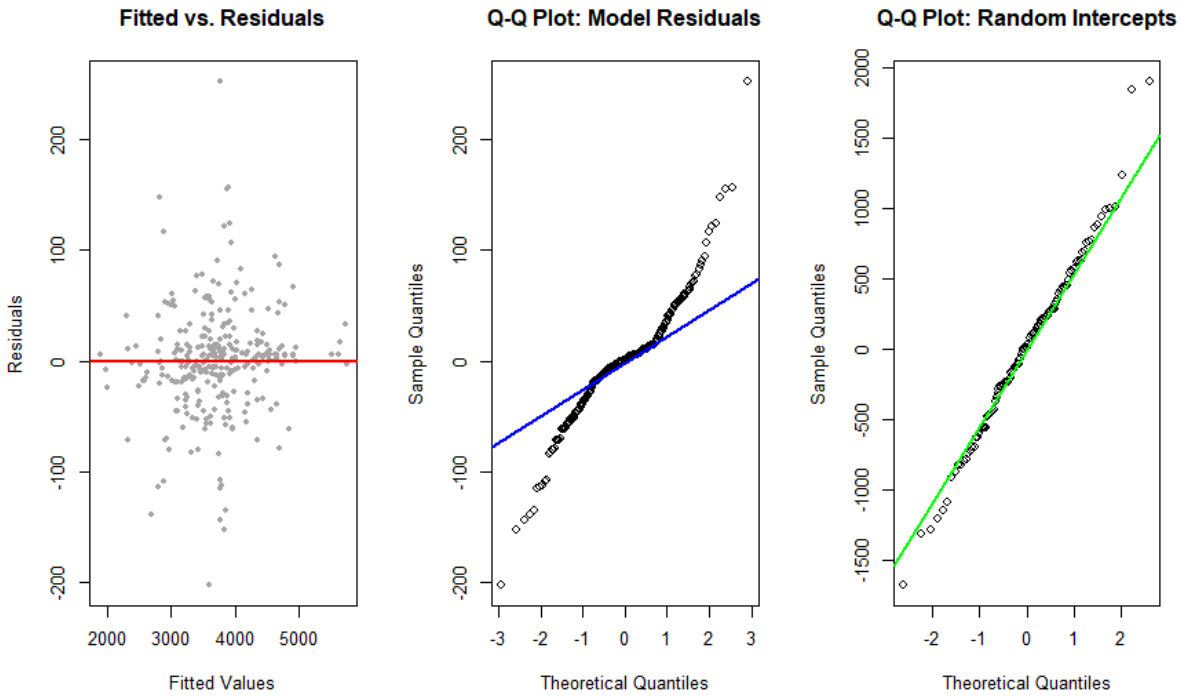
Table 5: Linear Model: Predicting Cognitive Score from Brain Volume

	Estimate	Std.Error	t-value	p-value
(Intercept)	39.5163	10.2462	3.8567	0.0002
HPC_L_TOT	-0.0015	0.0028	-0.5248	0.6003

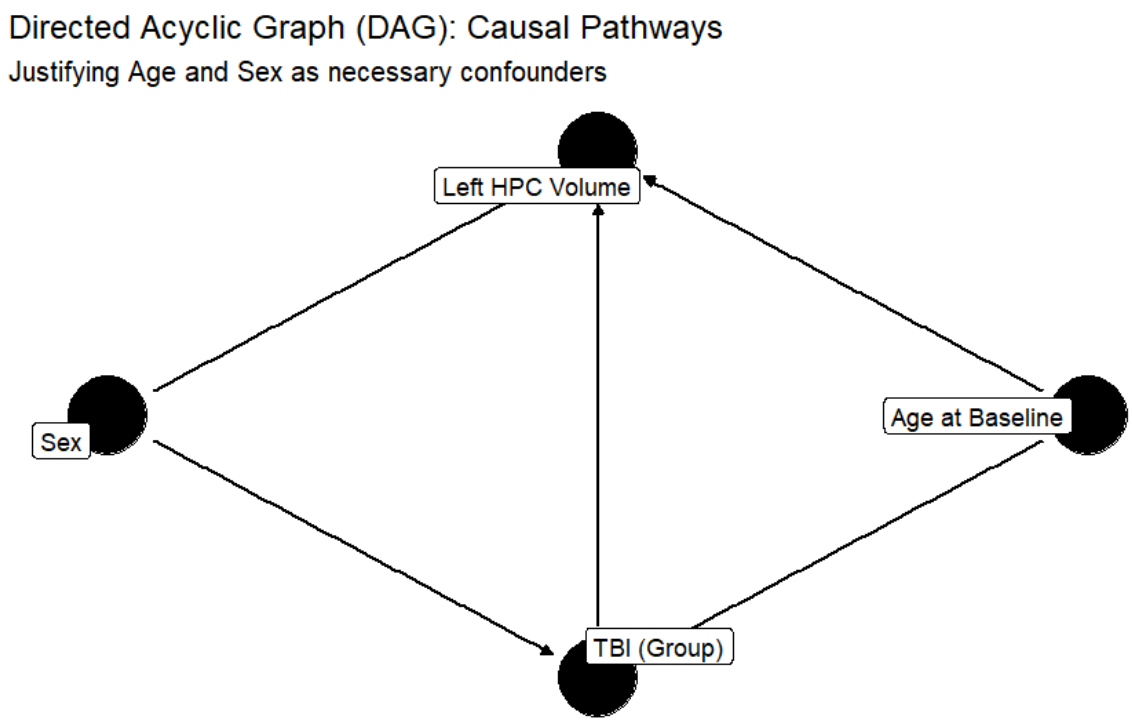


Model checking

Finally, the code produces several diagnostic plots for the adjusted mixed model, including fitted-versus-residual plots, a Q-Q plot of residuals, and a Q-Q plot of random intercepts. These are used to visually assess whether the linear mixed model assumptions are reasonably satisfied.

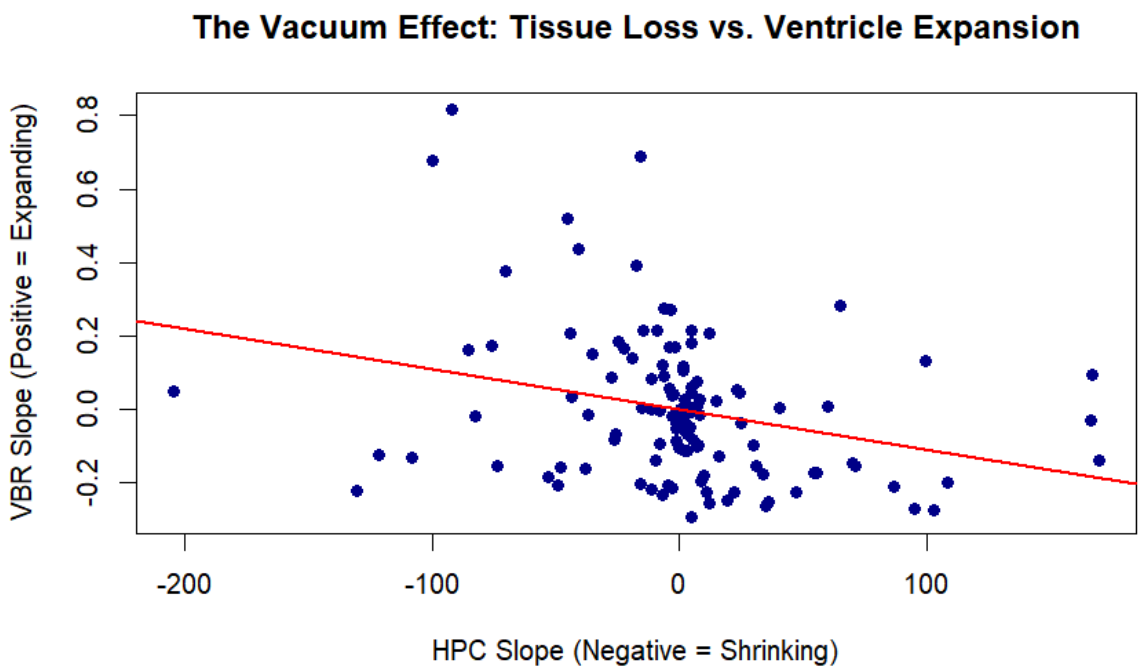


The DAG confirmed that we needed to include Age and Sex in our models as confounders. Age naturally shrinks brain volume over time, and sex influences baseline head size. Controlling them ensures we are isolating the actual damage caused by the TBI.



Validating the Decay: The Vacuum Effect To prove this tissue loss was real and not just measurement noise, we looked at the Ventricular Brain Ratio (VBR). The brain sits in a closed skull; if tissue shrinks, cerebrospinal fluid has to expand to fill the empty space. This means that we would expect an inverse relationship of VBR and HPC volume. It serves as a secondary validation, which would confirm that the observed volume loss is genuinely from degeneration of the brain.

We extracted the slopes from our models and found a significant negative correlation ($r = -0.287$, $p = 0.0017$). As hippocampal tissue goes down, ventricular expansion goes up, perfectly validating our structural findings.



Limitations

Although satisfied with the results, we found two main limiting factors to our analysis.

First, one significant limitation was the constraint of sample size. A limited number of participants (125 injured and 30 controls) could have restricted the power of a study. It is even more important to note that the portion of control to TBI patients is very skewed. This meant that any little change in the control data could have severely affected the results. Ideally, we would have an equal portion of control to TBI patients to adjust for more

variation and outliers. This could have made it difficult to detect small but significant effects. This means that the findings in our analysis could apply less to the broader population, limiting the results.

Second, although we did find a clear relationship (TBI patients had clear degeneration of the brain tissue over time) the scope of a project naturally excludes a wide range of important variables. These can include but are not limited to: environmental factors, treatments, or pre-existing conditions that were not measured during the study. These variables could interact with the primary variables in ways that the analysis is unable to capture. This highlights that the findings are focused into a specific scope rather than a full analysis of the injured patients over time.

Conclusion

We can see how Traumatic Brain Injury (TBI) impacts patients in the long run. Linear mixed effects model analysis shows that TBI is not some sort of isolated event in the whole brain but causes chronic, rapid brain degeneration. In fact, patients with TBI experience a decline of 65mm^3 of the left hippocampal volume per year compared to healthy brain tissue. This brain tissue loss was significantly inverse correlation ($r = -0.287$) to the VBR and clearly indicates that the brain shrinkage is a result of structural degeneration. When we analyzed cognitive therapy and the effects on structural decline, we found a slight positive trend which indicated that cognitive therapy protects brain tissue. However, we could not find strong evidence that the increased therapy hours actually helped to slow down volume loss. In addition, we found no correlation between the preserved hippocampal volume and the patients' LAQ scores. In other words, we have confirmed that greater brain volume doesn’t necessarily translate into increased cognitive capabilities. This is a strong indication that recovery from TBI is based less on volume preservation. Unfortunately, we had to take into account the unbalanced sample size (125 patients and 30 participants) which can be at variance with the general population. There are also likely more confounding factors such as environmental factors, treatments, and pre-existing conditions not considered in this study. Future study should involve larger and more balanced groups and address the clinical variability to understand more clearly the link between structural damage and cognitive recovery. This study has shown that treatments of TBI are more than just physical brain preservation but rather a full-fledged treatment plan that takes into account the patient’s

actual cognitive functioning in day-to-day cognition and allows for functional recovery beyond structural limitations.